

Amogh Vinaykumar

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EDUCATION

Flower Mound High School

Flower Mound, TX

Incoming Senior

Class of 2027

Outside Courses: Linear Algebra (current), Machine Learning Specialization (Coursera), CS231n (Stanford), Intro to Synthetic Aperture Radar (NASA), CS50P (Harvard), Intro to Algorithms (MIT)

EXPERIENCE

Founder

October 2025 – Present

Altis

Flower Mound, TX

- Founded Altis (altisx.space), a synthetic aperture radar (SAR) R&D startup developing ML-driven geospatial intelligence software for infrastructure and flood risk analysis (10k+ in pre-seed funding)
- Building an end-to-end SAR processing pipeline (Sentinel-1 VV/VH) for change detection, flood extent mapping, and physics-aware depth estimation at the property level
- Collaborating with researchers at NASA JPL/Caltech and the German Aerospace Center (DLR) on advanced SAR preprocessing, coherence analysis, and predictive modeling for flood-based disasters

Research Intern

September 2025 – Present

Massachusetts Institute of Technology (MIT)

Remote

- Developing Python-based simulation frameworks for multi-agent negotiation scenarios in nonlinear and constrained environments under Dr. Mark Klein
- Calculating utility functions, Pareto frontiers, social welfare, Nash product, and other metrics across diverse contract hypervolumes
- Experimenting with optimization algorithms (e.g., genetic algorithms, simulated annealing, etc) and strategic behaviors including information revelation, incentive compatibility, and lying
- Refining concession protocols and decision tree splits to minimize node depth and improve efficiency

Research Intern

December 2025 – Present

Princeton AI Lab

Remote

- Conducting research on multimodal modeling and reinforcement learning applications
- Built Any2Poster ([Paper Link](#)), a project that converts arbitrary multimodal inputs into structured posters for NeurIPS 2026
- Investigating vision-language-action systems capable of perception, reasoning, and decision-making in open environments

Research & Automation Intern

March 2026 – Present

Hemut (YC X25)

Remote

- Deploying deep research agents on GCP integrating Perplexity **sonar-deep-research** and GPT-4o with a claim-level fuzzy verification pipeline via FastAPI and Nginx
- Developing a \$5M+ GTM AI operating system for freight operators enabling efficient fleet coordination
- Designing intelligent trucking software systems for carrier onboarding, compliance workflows, and dispatch optimization

Machine Learning Research Intern

June 2025 – September 2025

Southern Methodist University

Dallas, TX

- Worked on an app called *Behavior Buddy* aiding speech recognition for children with autism using LLMs, transcription/diarization (AssemblyAI & Pyannote), and deep learning
- Helped build a digital twin of the campus that uses machine learning to simulate real-time responses to triggered events like temperature increases
- Developed predictive sensing technology with Lockheed Martin collaborators using 3D scanning, sensor fusion, and more

PROJECTS

Aerogel Heat Shield Research

February 2025 – April 2026

Independent Research / Mentored Project

- Researched polymer-reinforced silica and zirconia aerogels for spacecraft thermal shielding
- Collaborated with teams from Northwestern and Berkeley, and conducted experimental work at Rice and UNT Discovery Park
- Presented work at multiple conferences, with an upcoming publication, and secured funding from MIT for a more advanced follow-on project

Autonomous ML-Powered Disaster Relief Drone (Sentinel)

July 2025 – October 2025

Personal Project

- Designed a low-cost drone (grant-funded) with camera-based detection via ML for areas prone to natural disasters
- Developed design and prototype in Fusion; integrated autonomous flight plan in Python
- Created ML model combining CNN-based computer vision, reinforcement learning for route optimization, and sensor integration to enable autonomous drone supply delivery/person recognition

PUBLICATIONS & PRESENTATIONS

Publications

- [1] **A. Vinaykumar**, “Optimizing Aerogel-Based Materials for Enhanced Thermal Protection in Spacecraft Reentry Conditions,” *AIAA ASCEND*, Washington DC, 2026 (Accepted).
- [2] **A. Vinaykumar**, “ALTIS: Automated Loss Triage and Impact Scoring from Sentinel-1 SAR for Property-Level Flood Damage Assessment,” *arXiv*, <https://arxiv.org/abs/2603.13803>, 2026.

Presentations

- [1] **A. Vinaykumar**, “Sex-Dependent Skeletal and Neurobehavioral Responses to Combined Galactic Cosmic Radiation and Microgravity,” *American Society for Gravitational and Space Research (ASGSR) Annual Meeting*, 2025.
- [2] **A. Vinaykumar**, “Optimizing Aerogel-Based Materials for Enhanced Thermal Protection in Spacecraft Reentry Conditions,” *MIT Undergraduate Research Technology Conference (URTC)*, Cambridge, MA, 2025.
- [3] **A. Vinaykumar**, “Novel Composite Materials for Extreme Thermal Environments,” *American Chemical Society (ACS) Regional Symposium*, 2025.

AWARDS & HONORS

MIT THINK Finalist — 5/1042 (Aerogel heat shield research proposal) <i>Massachusetts Institute of Technology (MIT)</i>	2026
Conrad Challenge Finalist — 30/5000+ (Autonomous disaster prediction & delivery drone (<i>Team Altis</i>)) <i>Conrad Foundation</i>	2026
Texas Ideas Challenge Finalist — 40/600+ (Project Sentinel) <i>Texas A&M University</i>	2026

TECHNICAL SKILLS

Languages: Python, Java, Swift, MATLAB, Lisp, basic HTML/JavaScript
Frameworks/Libraries: PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, FastAPI
Platforms/Tools: Linux, AWS, Docker, COMSOL, Hugging Face, Fusion/Onshape, Supabase, Linear, HubSpot
Research Areas: Machine Learning, Computer Vision, Multimodal AI, Algorithmic Modeling